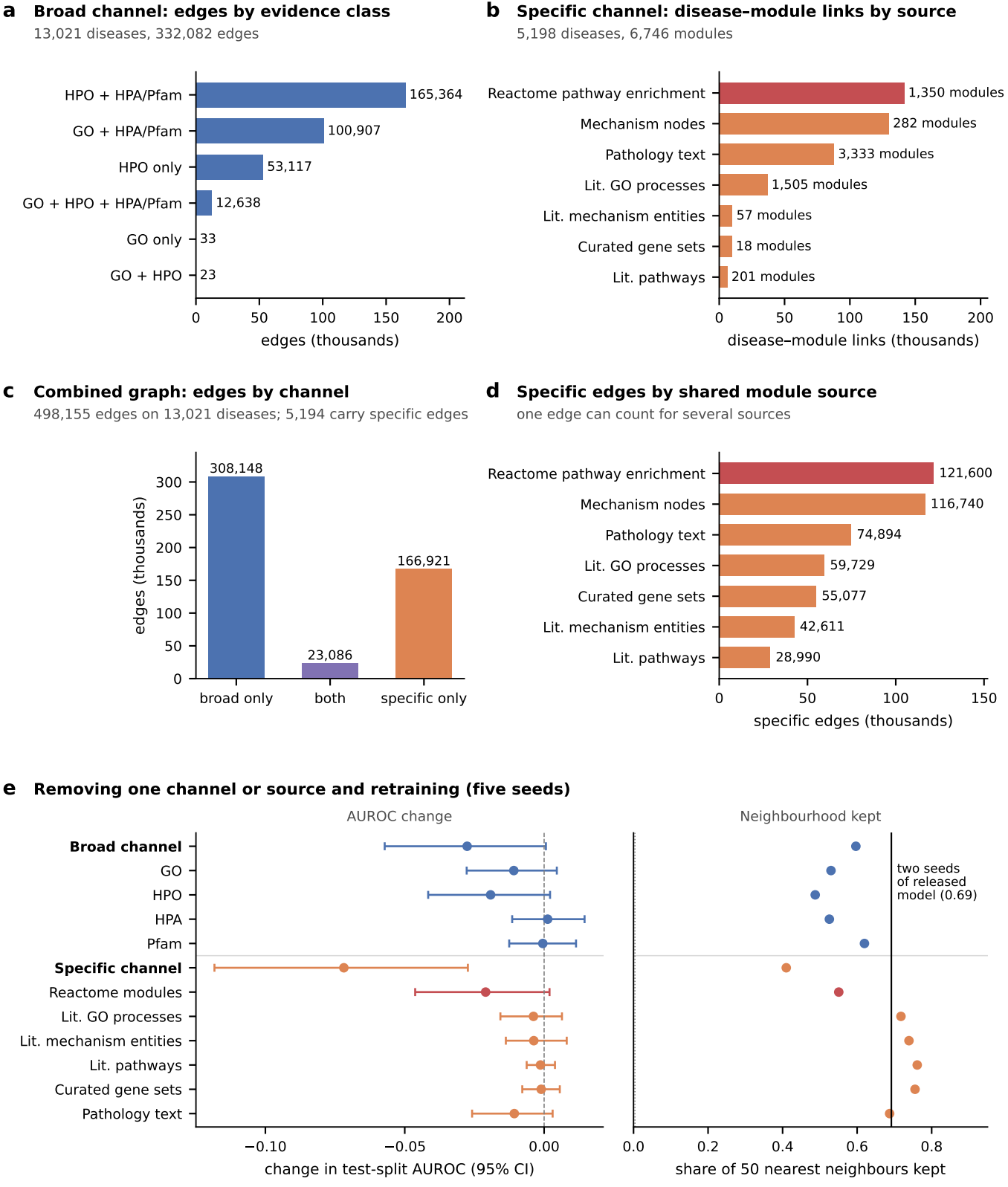
a
Broad channel: edges by evidence class
13,021 diseases, 332,082 edges
HPO + HPA/Pfam
GO + HPA/Pfam
HPO only
GO + HPO + HPA/Pfam
GO only
GO + HPO
165,364
100,907
53,117
12,638
33
23
0
50
100
150
200
edges (thousands)
b
Specific channel: disease–module links by source
5,198 diseases, 6,746 modules
Reactome pathway enrichment
Mechanism nodes
Pathology text
Lit. GO processes
Lit. mechanism entities
Curated gene sets
Lit. pathways
1,350 modules
282 modules
3,333 modules
1,505 modules
57 modules
18 modules
201 modules
0
50
100
150
200
disease–module links (thousands)
c
Combined graph: edges by channel
498,155 edges on 13,021 diseases; 5,194 carry specific edges
edges (thousands)
0
50
100
150
200
250
300
308,148
23,086
166,921
broad only
both
specific only
d
Specific edges by shared module source
one edge can count for several sources
Reactome pathway enrichment
Mechanism nodes
Pathology text
Lit. GO processes
Curated gene sets
Lit. mechanism entities
Lit. pathways
121,600
116,740
74,894
59,729
55,077
42,611
28,990
0
50
100
150
specific edges (thousands)
e
Removing one channel or source and retraining (five seeds)
AUROC change
Broad channel
GO
HPO
HPA
Pfam
Specific channel
Reactome modules
Lit. GO processes
Lit. mechanism entities
Lit. pathways
Curated gene sets
Pathology text
−0.10
−0.05
0.00
change in test-split AUROC (95% CI)
Neighbourhood kept
two seeds
of released
model (0.69)
0.0
0.2
0.4
0.6
0.8
share of 50 nearest neighbours kept